Risk Management: Technical Worksheet

Grade 11 · Worksheet · About 25 minutes
Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

> **How to use this worksheet**
> - Do the parts **in order** from top to bottom.
> - Each part has a **time** so you know how long to spend.
> - Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Risk management is a structured cycle: **identify** what could go wrong, **assess** how likely and how severe it is, then choose to **mitigate**, **transfer**, **avoid**, or **accept** the risk. You will practice separating likelihood from impact and documenting reasoned decisions for digital systems.

Do this:

1. **Name one digital risk** for a school or personal project (data loss, account takeover, outage before a deadline).

   _____

2. In a few words, is your concern mainly about **how often** it might happen, **how bad** it would be, or both?

   _____

Part 2 – Identify risks vs vague fears (about 5 minutes)

A useful risk statement names an **asset** (what you care about), a **threat** or hazard, and a possible **consequence**. “Something bad with the network” is too vague to manage. Clear identification enables assessment and controls.

> **Real-life scenario:** A robotics club stores competition code only on one laptop. Vague fear: “We might lose stuff.” Clear risk: “If the laptop disk fails before finals, we lose the only build and cannot compete.”

Discuss with a partner or reflect in writing: Why does rewriting vague fears into asset–threat–consequence statements help teams act?

Do this: Rewrite each vague statement as a clearer risk (asset + what could happen + consequence).

Vague statement	Clearer risk
“The cloud is risky.”	
“Phones are a problem.”	
“Wi-Fi might mess up.”	

Part 3 – Likelihood vs impact (about 5 minutes)

Likelihood is how probable an event is. **Impact** is how severe the harm would be if it occurs. High impact + low likelihood (rare catastrophic disk failure) and high likelihood + lower impact (frequent brief Wi-Fi blips) need different responses. Rough scales (Low / Medium / High) are enough for classroom analysis.

> **Real-life scenario:** Risk A: ransomware encrypts the student information system (low–medium likelihood for a well-patched school, very high impact). Risk B: a club printer jams weekly (high likelihood, low impact if a backup printer exists).

Critical thinking: Which risk deserves priority spending if the budget allows only one major control—and why? Use likelihood and impact in your answer.

Do this: Rate each risk **L** / **M** / **H** for likelihood and impact (your reasoned judgment).

Risk	Likelihood	Impact
Forgotten USB with unencrypted club passwords is lost on a bus	_____	_____
Brief Wi-Fi drop during advisory announcements	_____	_____
No tested backup of graduation photo archives before a disk failure	_____	_____

Part 4 – Treat risks: mitigate, transfer, avoid, accept (about 5 minutes)

After assessment, choose a treatment: **Mitigate** (reduce likelihood or impact with controls), **Transfer** (shift some consequence, e.g. insurance or a vendor SLA), **Avoid** (do not do the activity), or **Accept** (knowingly live with residual risk when cost of more control outweighs benefit). Acceptance should be deliberate—not accidental neglect.

> **Real-life scenario:** A journalism site collects tips. Mitigate: MFA and encryption. Avoid: refuse anonymous uploads entirely (hurts mission). Accept: residual risk that a determined attacker still breaches despite controls—documented and reviewed yearly.

Do this: Choose the **best-fit treatment** for each case (Mitigate / Transfer / Avoid / Accept).

Situation	Treatment
Store competition code on one laptop; add cloud + offline backups and weekly restore tests	_____
Cancel a public live stream of unfinished exam questions after spotting a leak risk	_____
Buy device insurance for loaner laptops taken on field trips	_____
Keep using a popular free plugin with a known low-severity bug; document why and revisit next term	_____

Write one residual risk you would still accept after mitigating the robotics-laptop scenario (what remains, and why acceptance is reasonable).

Part 5 – Risk register sketch (about 5 minutes)

Do this:

1. **Sketch a mini risk register** with four columns for one school digital service: Risk · Likelihood · Impact · Treatment. Fill at least two rows.

2. **Star the risk** you would treat first and justify using likelihood × impact thinking.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

☐ I can write a clear risk (asset, threat, consequence)—not only a vague fear.

☐ I can separate **likelihood** from **impact**.

☐ I can choose among **mitigate / transfer / avoid / accept**.

☐ I can explain deliberate acceptance of residual risk.

☐ I can sketch a simple risk register and prioritize a treatment.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Risk Management (Technical). Central topic: Risk management. Main technology area: Digital risk assessment and controls.